REVIEW: MULTIMODAL SENSOR FUSION

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- Multimodal Signal Process
- SOTA methods of multimodal sensor fusion for various navigation system
  - Visual-LiDAR fusion algorithm
- Multimodal Sensor Fusion for detection/recognition
- Simulation: Camera and LiDAR
- Future works

Multimodal Signal Process[1~3]

- Modality: **Formats** of various types of data that collect information about phenomena of interest from systems in various fields (point-cloud, RGB, etc.)

Scientific Advancement → **Diversification of data needed in various fields** → **Signals with a wide variety of modalities are acquired**

Increasing demand for multimodal signal process research to handle signals with various modalities

➔ *Requires efficient and intuitive acquisition and processing of multimodal data*

Multimodal data representation

➔ ***What data*** *will be fused and expressed?*

Multimodal data Information fusion

➔ ***Convergence method*** *required to express multimodal data*

Multimodal Signal Process [1~3]

Multimodal Signal Process

Data Representation

LiDAR + **Camera**

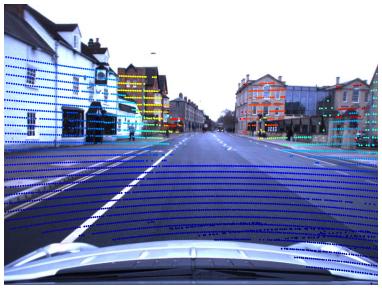


Fig. 1. Fusion of Camera and LIDAR

Information Fusion

	Learning	Navigation Filter
Early Fusion	*Combining at pre-learning stage*	*Convergence signal processing (RGBDP, etc.) for each raw data*
Late Fusion	*Combine after learning each*	*Calculate and fuse the necessary information from each raw data (GPS/INS, etc.)*
Intermediate Fusion	*Combining trained data and raw data*	*Fusion of calculated information and raw data (LiDAR odometry, etc.)*

SOTA METHODS OF MULTIMODAL SENSOR FUSION FOR VARIOUS NAVIGATION SYSTEM[4]

Visual-inertial fusion algorithms

- **MSCKF 2.0**[5]
  - ✓ Multi-state constraint KF
  - ✓ Monocular camera + IMU
- **ROVIO**[6]
  - ✓ Extended KF-SLAM
  - ✓ Monocular camera + IMU
- **VINS-Mono**[7]
  - ✓ {Sliding window + Pose graph} optimization
  - ✓ Monocular camera + IMU

Visual-LiDAR fusion algorithms

- **DEMO**[15]
  - ✓ Bundle adjustment
  - ✓ Mechanical LiDAR + RGB camera
- **LIMO**[16]
  - ✓ Bundle adjustment
  - ✓ Mechanical LiDAR + Mono camera
- **TVL-SLAM**[17]
  - ✓ Bundle adjustment
  - ✓ Mechanical LiDAR + Mono camera
- **Method**[18]
  - ✓ Pose graph optimization
  - ✓ Mechanical LiDAR + Mono camera

LiDAR-inertial fusion algorithms

- **LIOM**[8]
  - ✓ Sliding window optimization
  - ✓ Mechanical LiDAR + IMU
- **LION**[9]
  - ✓ Sliding window optimization
  - ✓ Mechanical LiDAR + IMU
- **LINS**[10]
  - ✓ Iterated error-state KF
  - ✓ Mechanical LiDAR + IMU
- **LIO-SAM**[11]
  - ✓ Factor graph optimization
  - ✓ Mechanical LiDAR + IMU
- **LILI-OM**[12]
  - ✓ {Sliding window + Pose graph + Factor graph}
  - ✓ Solid-state or Mechanical LiDAR + IMU
- **FAST-LIO**[13]
  - ✓ Iterated error-state KF
  - ✓ Solid-state LiDAR + IMU
- **Faster-LIO**[14]
  - ✓ Iterated error-state KF
  - ✓ Solid-state LiDAR + IMU

LiDAR-visual-inertial fusion algorithms

- **LIC-Fusion 2.0**[19]
  - ✓ Multi-state constraint KF
  - ✓ Mechanical LiDAR + IMU + Mono camera
- **Super odometry**[20]
  - ✓ Factor graph optimization
  - ✓ Mechanical LiDAR + IMU + Mono camera
- **LVI-SAM**[21]
  - ✓ Factor graph optimization
  - ✓ Mechanical LiDAR + IMU + Mono camera
- **R3LIVE**[22]
  - ✓ Iterated error-state KF
  - ✓ Mechanical LiDAR + IMU + Mono camera

VISUAL-LIDAR FUSION ALGORITHM

- Lidar-Monocular Visual Odometry using Point and Line Features (LIMO)[16]
  - Target: LiDAR-monocular visual odometry pose estimation accuracy improvement
    - Lidar-monocular visual odometry method combining both point and line features in a purely geometric
      - Monocular camera-based point and line detection
      - Extract the depth of detected points and lines by matching 3D point-cloud to RGB image
      - Using the above information, estimate the vehicle's pose.

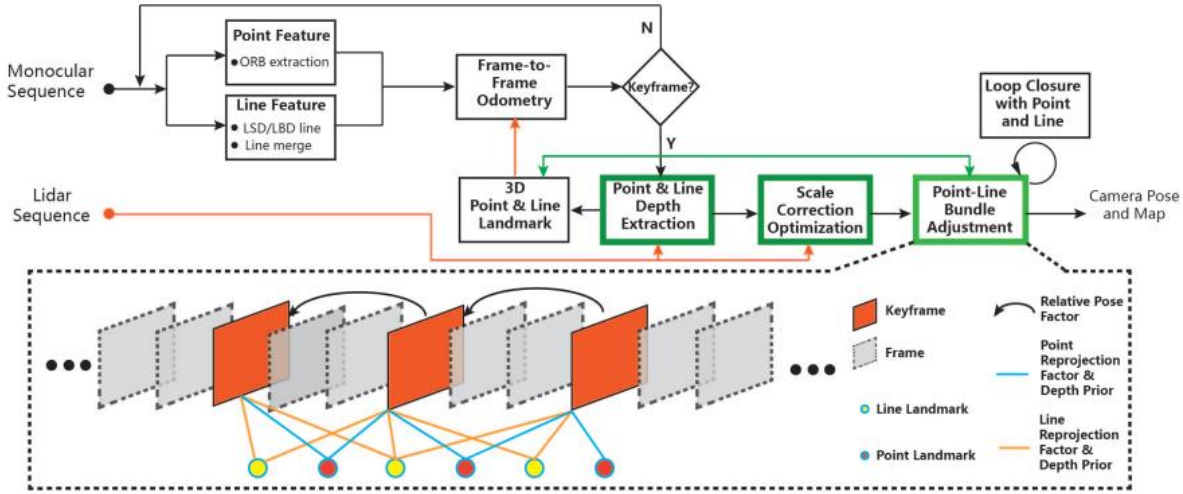


Fig. 2. Structure of LIMO

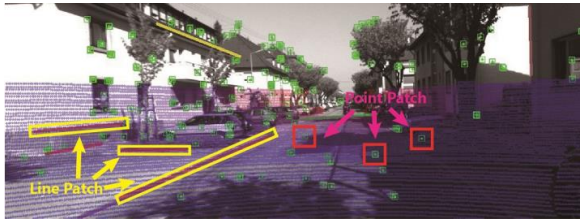


Fig. 3. Point and Line depth extraction

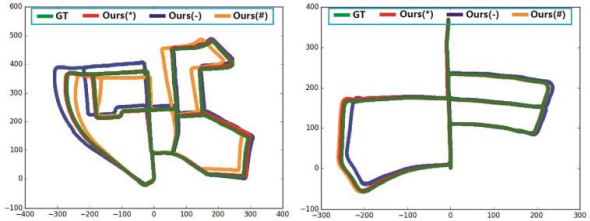


Fig. 4. Pose estimation results

Multimodal Sensor Fusion for Detection/Recognition

- Fusion of 3D LiDAR and camera for object detection
- BEVDepth: Acquisition of reliable depth for multi-view 3D object detection

FUSION OF 3D LIDAR AND CAMERA FOR OBJECT DETECTION[23]

- Target: Obtain highly accurate location and classification of objects in the driving environment
  - Multi-object detection methodology
    - Generation of 3D LiDAR-based object area proposals in point-cloud
    - Convergence of RGB image with 3D point-cloud and LiDAR-based object-region
    - Multi-scale CNN (Convolutional Neural Network)-based object detection

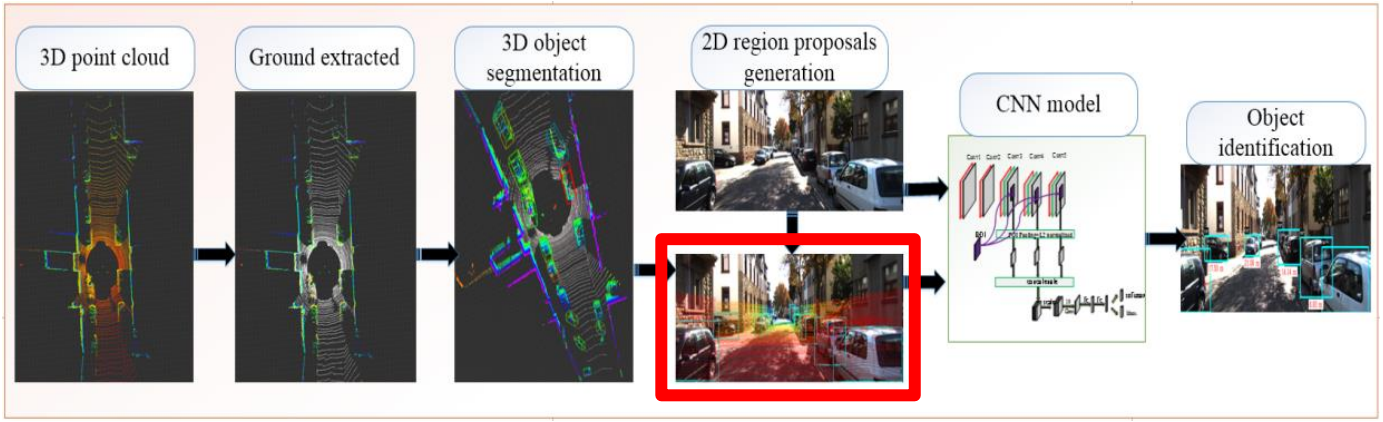


Fig. 4. Structure of multi-object detection methodology

- ***{Intermediate} fusion methodology***
- ***Limitation***
  - ✓ ***In the fusion situation of 3D LiDAR and Camera, fusion occurs without considering the difference in resolution***
  - ✓ ***The quality (accuracy) of converged information may be reduced***

BEVDEPTH: ACQUISITION OF RELIABLE DEPTH FOR MULTI-VIEW 3D OBJECT DETECTION[24]

- Target: Camera-based Bird's-Eye-View (BEV) 3D object detection
  - A new 3D object detector with reliable depth estimation
    - To guide depth learning, information that combines 3D point-cloud and RGB is utilized

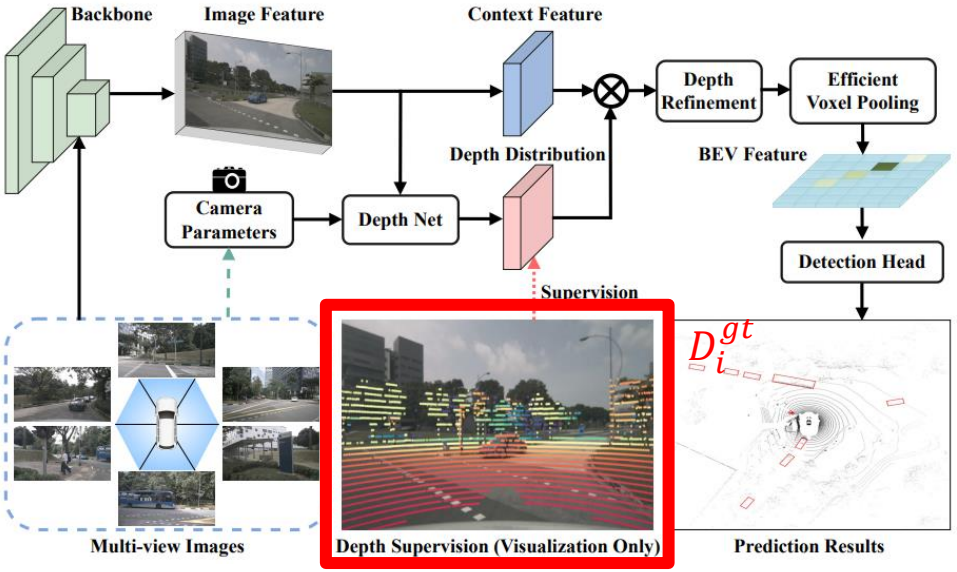


Fig. 5. Structure of Camera-based BEV 3D object detection

< Mapping 3D point-cloud data to RGB image >

$$P_i^{img}(u, v, d) = K_i(R_i P + t_i)$$

K: Intrinsic matrix *R: Rotation matrix*
P: LiDAR point-cloud *t: Translation matrix*

< Obtain depth supervision >

$$D_i^{gt} = \phi\left(P_i^{img}\right)$$

ϕ: ① *Minimum Pooling*
→*Extract closest distance information*
② *One hot*
→*Apply the result of ① to the depth dimension*

➢ ***Limitation***
- ✓ ***Overlap of depth information caused by resolution difference in 3D LiDAR and camera fusion situation → Solved with Minimum Pooling***
- ✓ ***Possibility of forming inaccurate depth supervision***

Simulation: Camera and LiDAR

- Sensor setup for data matching
- Geometric alignment process
- Simulation configuration
- Data of sensors
- Projection results

SENSOR SETUP FOR DATA MATCHING[25]

- For fusing LiDAR, RADAR, and stereo camera sensors, a **geometric alignment** process is required to **convert it into a data format with one unified frame**.

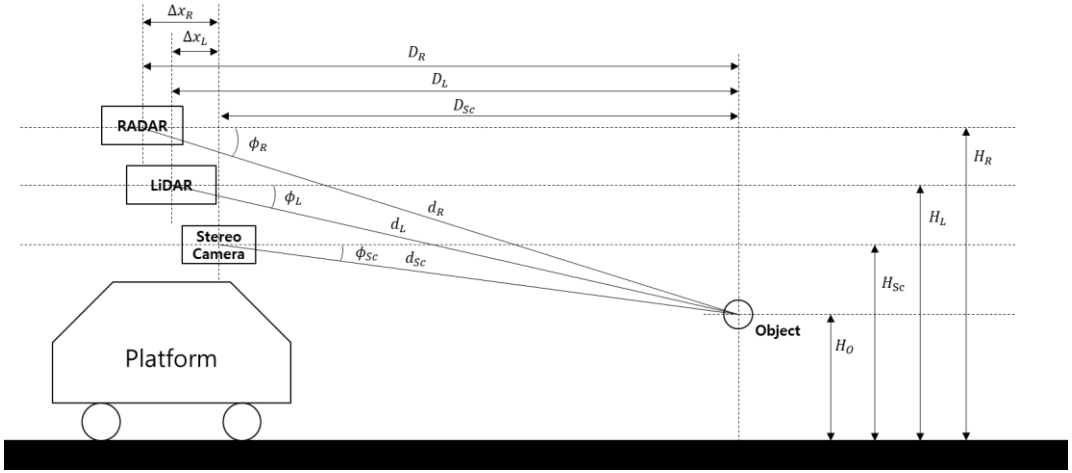


Fig. 6. Side view of sensor setup

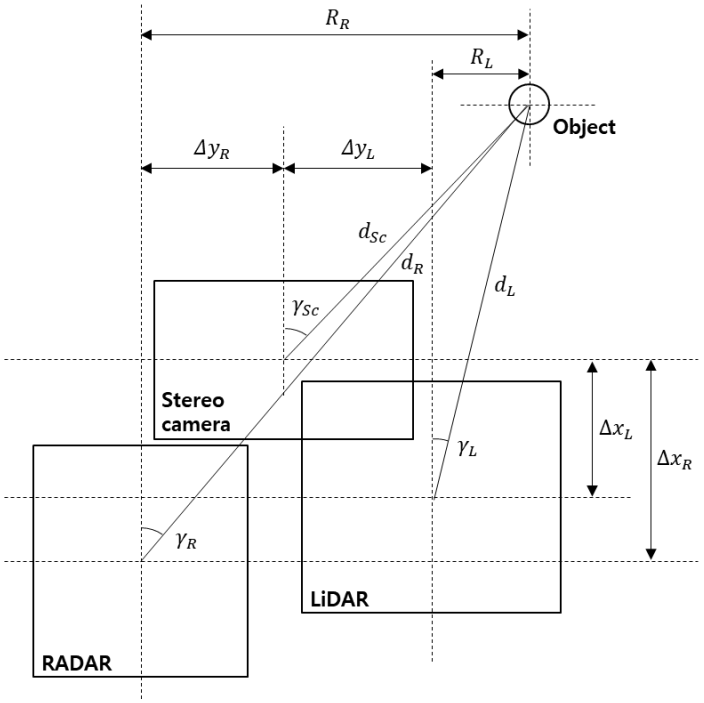


Fig. 7. Top view of sensor setup

- ✓ **Criteria for Geometric Alignment: RGB image (Stereo camera)**
- ✓ LiDAR data: $p_L = [\phi_L \quad \gamma_L \quad d_L]^T$
- ✓ RADAR data: $p_R = [\phi_R \quad \gamma_R \quad d_R]^T$
- ✓ Stereo camera data: $RGBD_{Sc} = [R_{Sc} \quad G_{Sc} \quad B_{Sc} \quad D_{Sc}]^T$
- ✓ ϕ, γ : latitude and longitude of object relative to sensor
- ✓ d : distance of object relative to sensor
- ✓ $\Delta x, \Delta y$: separation distance of the sensor from the stereo camera

GEOMETRIC ALIGNMENT PROCESS[25]

Forming a relationship

- ✓ D : vertical distance from sensor to object
  - → $D_L = d_L \cos\phi_L \cos\gamma_L = d_{Sc} \cos\phi_{Sc} \cos\gamma_{Sc} + \Delta x_L, D_R = d_R \cos\phi_R \cos\gamma_R = d_{Sc} \cos\phi_{Sc} \cos\gamma_{Sc} + \Delta x_R$ … (3)
- ✓ R : horizontal distance from sensor to object
  - → $R_L = d_L \cos\phi_L \sin\gamma_L = d_{Sc} \cos\phi_{Sc} \sin\gamma_{Sc} - \Delta y_L, R_R = d_R \cos\phi_R \sin\gamma_R = d_{Sc} \cos\phi_{Sc} \sin\gamma_{Sc} + \Delta y_R$ … (4)
- ✓ H : height of object
  - → $H_O = H_L - d_L \sin\phi_L = H_{Sc} - d_{Sc} \sin\phi_{Sc}, H_O = H_R - d_R \sin\phi_R = H_{Sc} - d_{Sc} \sin\phi_{Sc}$ … (5)

LiDAR data in stereo camera frame

$$\boldsymbol{\phi_L^{Sc}} = \tan^{-1} \frac{\left((H_{Sc} - H_L) + d_L \sin\phi_L\right) \cos\gamma_{Sc}}{d_L \cos\phi_L \cos\gamma_L - \Delta x_L}$$

$$\boldsymbol{\gamma_L^{Sc}} = \tan^{-1} \frac{d_L \cos\phi_L \sin\gamma_L + \Delta y_L}{d_L \cos\phi_L \cos\gamma_L - \Delta x_L}$$

$$\boldsymbol{d_L^{Sc}} = \frac{d_L \cos\phi_L \cos\gamma_L - \Delta x_L}{\cos\phi_L^{Sc} \cos\gamma_L^{Sc}}$$

… (6)

RADAR data in stereo camera frame

$$\boldsymbol{\phi_R^{Sc}} = \tan^{-1} \frac{\left((H_{Sc} - H_R) + d_R \sin\phi_R\right) \cos\gamma_{Sc}}{d_R \cos\phi_R \cos\gamma_R - \Delta x_R}$$

$$\boldsymbol{\gamma_R^{Sc}} = \tan^{-1} \frac{d_R \cos\phi_R \sin\gamma_R + \Delta y_R}{d_R \cos\phi_R \cos\gamma_R - \Delta x_R}$$

$$\boldsymbol{d_R^{Sc}} = \frac{d_R \cos\phi_R \cos\gamma_R - \Delta x_R}{\cos\phi_R^{Sc} \cos\gamma_R^{Sc}}$$

… (7)

SIMULATION CONFIGURATION

- Oxford RobotCar Dataset[26,27] (world-widely used open dataset)

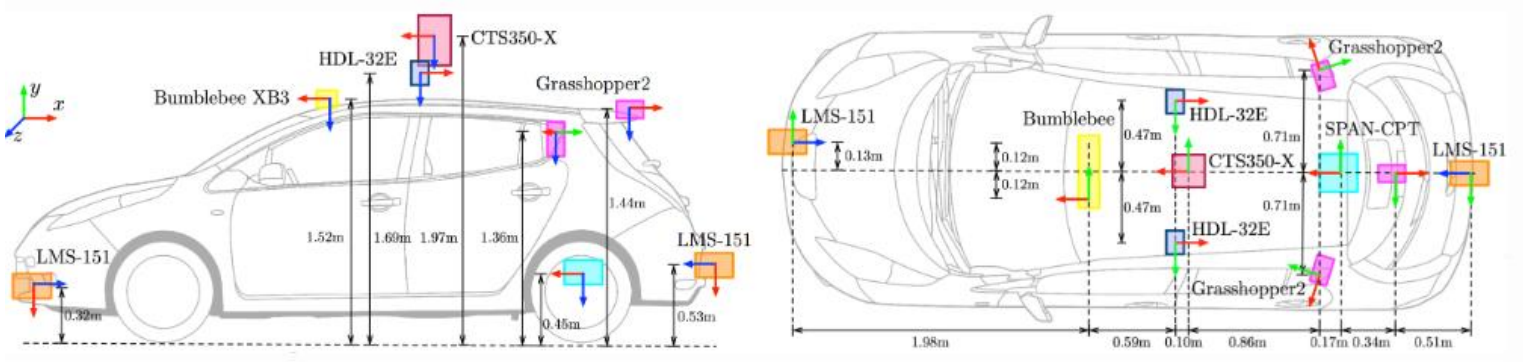


Fig. 8. Sensor setup – Oxford RobotCar Dataset

Table 1. Sensor specification

3D LiDAR		2D RADAR		Stereo camera	
Product name	HDL-32E (Velodyne)	Product name	CTS350-X (Navtech)	Product name	Bumblebee XB3 (Point Grey)
HFoV VFoV	360° 41.3°	Sampling rate	4 Hz	Resolution	1280 x 960 x 3
Sampling rate	20 Hz	Maximum range	163 m	Sampling rate	16 Hz
Maximum range	100 m	Range resolution	4.38 cm	Shutter	Global
Range resolution	2 cm	Beamwidth	1.8°	HFoV	66°
				Baseline	12/24 mm
				Lens focal length	3.8 mm

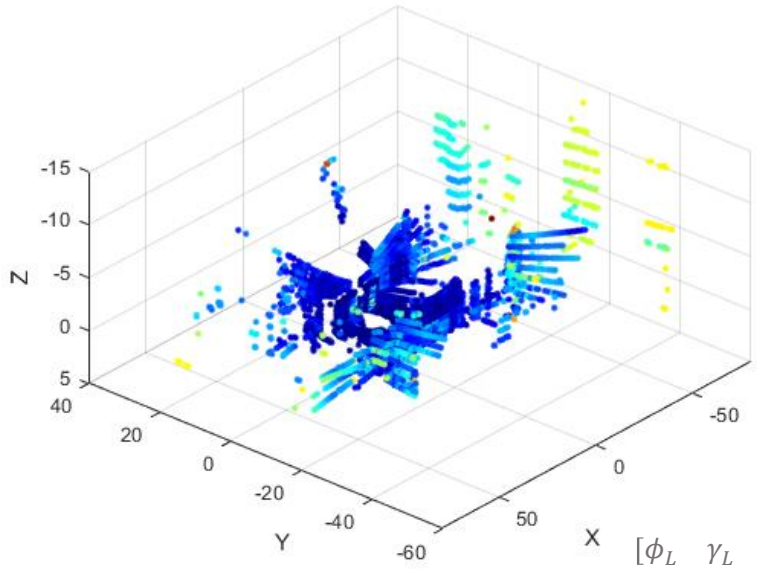
$H_{Sc} \approx 1.52$ m
$H_L \approx 1.69$ m
$H_R \approx 1.97$ m
$\Delta x_L \approx 0.59$ m
$\Delta x_R \approx 0.69$ m
$\Delta y_L \approx 0.47$ m
$\Delta y_R \approx 0$ m

Known Parameters

DATA OF SENSORS



Fig. 9. RGB image



$[\phi_L \quad \gamma_L \quad d_L \quad P_L]^T$

Fig. 10. LiDAR point cloud

PROJECTION RESULTS

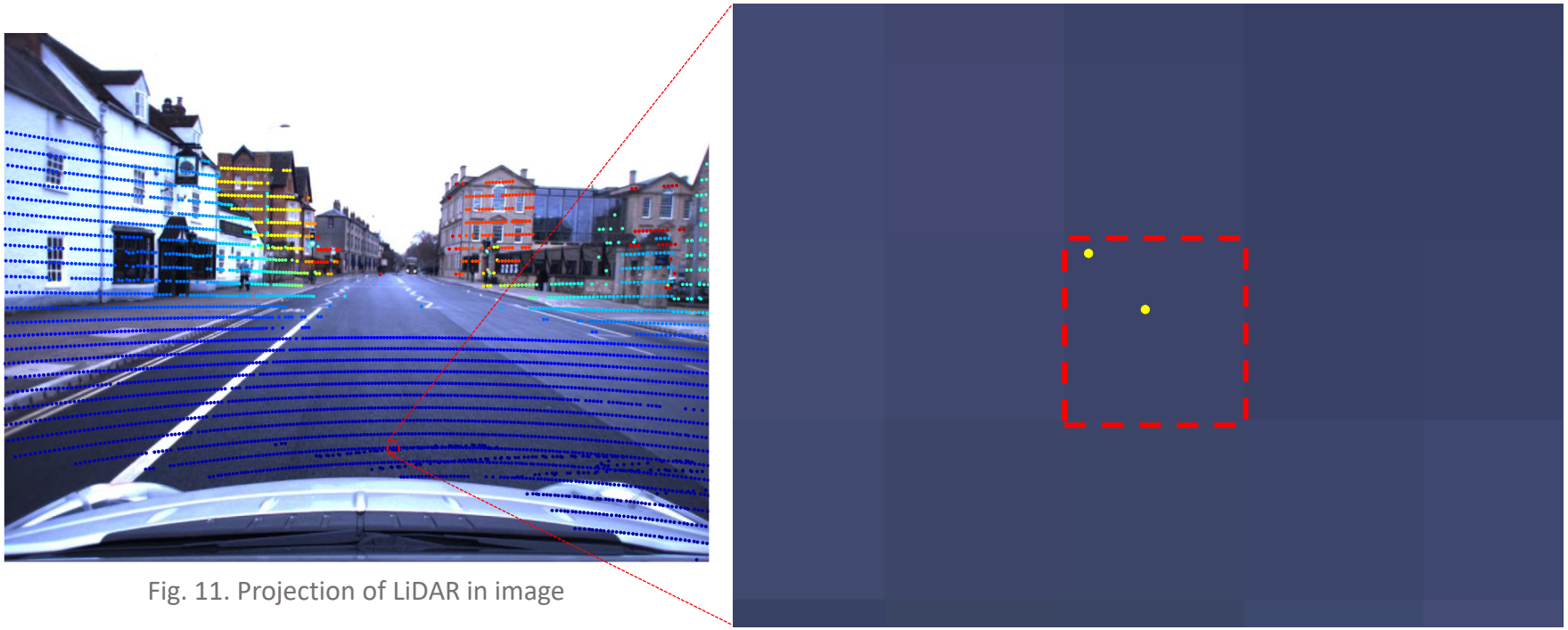


Fig. 11. Projection of LiDAR in image

- *Projection information: {1 frame of image} + {point-cloud stored for 0.05 sec}*
- *Even though LiDAR data was projected on the image for only 0.05 seconds (approximately 1 sweep),* ***it can be seen that 2 depth information were captured within 1 pixel***
- *In actual multimodal sensor fusion,* ***LiDAR data that has been swept multiple times is projected, so more point-cloud data will overlap***

Future Works

- **Early fusion of multimodal sensor**
  - Issue 1) Point-cloud data **overlap phenomenon** caused by **multiresolution**
  - Issue 2) **Interpolation technique** for 3D LiDAR point-cloud data with **sparse characteristics**
  - Issue 3) Performance improvement through LiDAR and (mono or stereo) camera **fusion in harsh environments** (improved depth data accuracy)

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THANK YOU

